APERIO | AUTONOMOUS EARTH INTELLIGENCE PLATFORM

# AUTHORITY DISASTER INTELLIGENCE ASSESSMENT • LANDSLIDE

Location: Meppadi, Wayanad, Kerala, India • Generated: 09 Oct 2026 03:50 UTC • Status: APPROVED BY OPERATOR

| | | | |
|---|---|---|---|
| **Target Event** | Landslide • Meppadi, Wayanad | **Risk Probability** | **91% (CRITICAL)** |
| **Affected Area** | 7.03 km² | **Model Confidence** | 95% Indicative |
| **Exposed Population** | 4,250 | **Antecedent Rainfall** | 180 mm (72h: 572mm) |
| **Terrain Slope** | 35° (Steep escarpment) | **SAM Scar Boundary** | Validated (7.03 km²) |
| **Satellite Observation** | Sentinel-1 SAR + Sentinel-2 MSI | **Human Verification** | APPROVED BY OPERATOR |

## Multi-Agent Intelligence & Evidence Progression

**1. Satellite Ingestion:** Copernicus Sentinel-1 C-SAR IW mode acquired observation passes over Meppadi, Wayanad. Radar backscatter penetration enabled continuous surface change mapping.
**2. DEM Terrain Modeling:** Horn algorithm computed local terrain slope of 35° across drainage lines.
**3. SAM Spatial Segmentation:** Delineated spatial impact boundary encompassing high-risk terrain.
**4. XGBoost & Bayesian Sequential Update:** Domain evidence likelihoods updated sequential prior belief to yield calibrated posterior risk of **91%**.

## Response Recommendations & Relocation Planning

| Designation | Capacity / Occupancy | Distance / Road Status | Hazard Exposure |
|---|---|---|---|
| Designated Relief Center (Meppadi, Wayanad) | 650 / 380 (270 Available) | 2.4 km • Open | **LOW RISK** |
| Community Relief Hall West | 420 / 310 (110 Available) | 4.1 km • Standby | **MODERATE** |
| District Camp North | 800 / 220 (580 Available) | 6.8 km • Clear | **LOW RISK** |

## Satellite Change Intelligence & Extent Quantification

| | | | |
|---|---|---|---|
| **Analyzed Extent** | 12056.04 km² (Full Granule) | **Methodology** | Visual Soil Exposure & Canopy Disturbance Delta |
| **Baseline Affected Extent** | 0.12 km² (2024-06-29) | **Sensor & Product** | Sentinel-2 MSI Level-2A (Tile T43PEN) |
| **Latest Affected Extent** | 781.72 km² (2024-08-13) | **Cloud Cover** | Pre: 59.04% • Post: 27.91% |
| **Newly Affected Area** | **781.6 km²** | **Net Change** | **781.6 km²** (+651333.3%) |
| **Detection Zones** | 6 Delineated Zones | **Integrity Status** | **CANDIDATE_CHANGE** |

## Accountable Operator Review & Audit Trail

**Reviewing Desk:** APERIO Response Operations Desk
**Operator Notes:** Automated verification report
**Operational Boundary:** Decision Support Output only. All evacuations require local ground confirmation.